Software Test Plan

Blood Bank Management System

Concise verification plan based on SRS Version 1.0

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Item	Details
Project	Blood Bank Management System
Source	BBMS Software Requirements Specification v1.0
Status	Draft; test execution not yet recorded

1.Introduction

This plan defines how we will verify the Blood Bank Management System. It covers functional workflows, performance, reliability, usability, maintainability, and security requirements from the SRS. All cases begin with status N because no execution evidence has been supplied.

2.Test Items

- Authentication and role-based access.
- Donor, eligibility, donation and history functions.
- Inventory availability, search, status and expiry.
- Recipient, request, allocation and issue workflows.
- Admin user management, reports and audit trail.
- Web UI, REST API and relational database integration.

3.Features to be Tested

Requirement	Feature	Priority	Test Cases
BBMS-F-001 to F-004	Donor registration, updates, eligibility and donation history	High	TC-DON-01 to 04
BBMS-F-010 to F-014	Donation recording, inventory creation, search and expiry	High	TC-INV-01 to 04
BBMS-F-020 to F-027	Recipients, requests, status, allocation and issue	High	TC-REC-01; TC-REQ-01 to 03; TC-ALLOC-01; TC-ISSUE-01
BBMS-F-030 to F-036	Authentication, sessions, RBAC, admin, reports and audit	High	TC-AUTH-01 to 03; TC-ADMIN-01 to 03
BBMS-NF-001 to NF-007	Performance, load, availability, security, UX, recovery and maintainability	High / Medium	TC-PERF-01; TC-REL-01 to 02; TC-UX-01; TC-MAINT-01
BBMS-SR-001 to SR-005	TLS, password hashing, authorization, input safety and audit	High / Medium	TC-SEC-01 to 05

4.Features Not to be Tested

- Clinical diagnosis, treatment decisions, and medical judgement.
- Internal behavior of external hospital or laboratory systems.
- Production-scale national deployment, payment gateways, and hardware device firmware.
- Donor or recipient self-service accounts unless the SRS is revised to include those actors.

5.Test Approach and Strategy

5.1.Test Levels

- Unit tests for validators, rules, services, and repositories.
- API and database integration tests for transactions and authorization.
- System tests for complete staff and admin workflows.
- Acceptance tests against SRS criteria and the RTM.

5.2.Test Types

Functional, negative, boundary, regression, browser compatibility, performance, reliability, usability, and security testing will be used. Test data will be synthetic and will not contain real donor or recipient information.

5.3.Cross Testing Strategy

Module	Owner	Cross Tester
Donor and Donation	G S Retikka Prasad	Jyothika M
Inventory	Jyothika M	Gaganashree R
Recipient and Request	Gaganashree R	Durugesh Bhergi
Authentication Admin and Integration	Durugesh Bhergi	G S Retikka Prasad

5.4.Entry and Exit Criteria

Entry: reviewed requirements, deployable build, prepared environment, approved business-rule tables, and available test data. Exit: all planned cases executed, all high-priority requirements passed, no open critical defects, acceptable performance and security results, and an updated RTM.

6.Security Validation

Verify TLS, password hashing, backend RBAC, session invalidation, input validation, safe database access, least-privilege data exposure, safe error messages, audit logging, and login throttling.

7.Test Environment

Category	Environment
Client	Current Chrome, Firefox and Edge; desktop and mobile viewport.
Server	Node.js and Express test build with isolated configuration.
Database	PostgreSQL test database with resettable seeded records.
Tools	Jest, Supertest, Playwright, Postman or equivalent, and a load-test tool.
Data	Synthetic donors, recipients, users, donations, units, requests and boundary dates.

8.Test Schedule

Milestone	Timing	Output
Plan and case review	Week 1	Approved plan and mapped cases
Environment and smoke test	Week 1	Stable test environment
Module and API testing	Week 2	Results and defect records
Integration and regression	Week 3	End-to-end results
Performance security usability	Week 4	NFR and security evidence
Acceptance and closure	Final review	RTM and final test summary

9.Test Deliverables

Test Plan, mapped test cases, synthetic test data, automated scripts where practical, execution records, defect reports, RTM, performance and security evidence, and Final Test Summary Report.

10.Roles and Responsibilities

Name	Primary Responsibility
G S Retikka Prasad	Project coordination, donor module, RTM review, cross-test authentication and admin.
Jyothika M	Inventory and expiry tests; cross-test donor and donation.
Gaganashree R	Recipient, request, allocation and issue tests; cross-test inventory.
Durugesh Bhergi	Authentication, admin, reports, audit and integration; cross-test recipient and request.

11.Risks and Mitigation

Risk	Impact	Mitigation
Eligibility or compatibility rules not approved	Core tests may be ambiguous	Approve rule tables before relevant execution.
Unstable build or environment	Blocked or misleading results	Smoke-test every build and keep a resettable database.
Limited realistic data	Boundary cases may be missed	Use synthetic combinations, expiry dates and request states.
Concurrent allocation defects	Same unit may be assigned twice	Run parallel requests and verify locks and constraints.
Sensitive data exposure	Privacy and security failure	Use synthetic data and inspect permissions, logs and errors.

12.Assumptions and Dependencies

- Only Staff and Admin roles are tested unless the SRS is formally changed.
- The approved eligibility and compatibility rules are available before workflow testing.
- The test environment can be reset and contains no real personal or medical data.
- Supported browsers, Node.js services and PostgreSQL are available to the team.

13.Suspension and Resumption Criteria

Suspend testing when the environment is unavailable, a blocking defect prevents meaningful execution, the data state cannot be trusted, or business rules required for expected results are unresolved. Resume after the blocker is fixed, the build passes smoke testing, and the database is restored to a known state.

14.Test Case Management and Traceability

Status codes: N - Not Run, P - Passed, F - Failed, B - Blocked. Detailed evidence and defect links are added during execution.

TC ID	Req	Test Procedure	Expected Result	Status
TC-DON-01	F-001	Create a donor with all mandatory valid data.	A unique donor record is saved.	N
TC-DON-02	F-002	View and update a donor as authorized staff; repeat as unauthorized user.	Permitted changes persist; unauthorized access is denied.	N
TC-DON-03	F-003	Enter valid and invalid eligibility information.	Eligibility is stored when valid and rejected with feedback when invalid.	N
TC-DON-04	F-004	Record multiple donations and open donor history.	Only donations linked to that donor are displayed.	N
TC-INV-01	F-010	Record a donation with donor, date, group, quantity and status.	The complete donation record is stored.	N
TC-INV-02	F-011	Complete a valid donation and inspect inventory.	Corresponding inventory units are created or updated once.	N
TC-INV-03	F-012, F-013	Filter inventory by group and status and compare with seeded records.	Displayed and filtered stock matches stored data.	N
TC-INV-04	F-014	Test units before, on, and after the expiry boundary.	Expired units are identified and excluded from available stock.	N
TC-REC-01	F-020, F-021	Create, search, view and update a recipient record.	Authorized operations return the correct unique recipient.	N
TC-REQ-01	F-022, F-023	Submit complete, incomplete and rule-invalid blood requests.	Valid requests receive IDs; invalid requests show field-level errors.	N
TC-REQ-02	F-024	Attempt allowed and disallowed request-state transitions.	Allowed transitions persist; invalid transitions are rejected.	N
TC-REQ-03	F-025	Process a request with permitted and non-permitted roles.	Only authorized staff can process the request.	N
TC-ALLOC-01	F-026	Allocate compatible stock to an approved request, including insufficient and concurrent cases.	No allocation exceeds or double-reserves available stock.	N
TC-ISSUE-01	F-027	Issue a valid allocation and attempt to issue it again.	Issue is recorded once; unit and request statuses update atomically.	N
TC-AUTH-01	F-030	Log in with valid and invalid credentials, including an unauthorized account.	Valid access succeeds; invalid access is denied without data leakage.	N

14.Test Cases Continued

TC ID	Req	Test Procedure	Expected Result	Status
TC-AUTH-02	F-031, F-033	Open protected staff and admin functions using each role.	Backend RBAC allows only the assigned role.	N
TC-AUTH-03	F-032	Expire or invalidate a session and call a protected endpoint.	Access is denied and re-authentication is required.	N
TC-ADMIN-01	F-034	Create, disable and change the role of a user account.	Changes take effect without bypassing admin authorization.	N
TC-ADMIN-02	F-035	Generate reports from known seeded donations, stock, requests and issues.	Report values match source records and filters.	N
TC-ADMIN-03	F-036	Perform representative admin actions and inspect the audit trail.	User, action and timestamp are recorded.	N
TC-PERF-01	NF-001, NF-002	Run the defined test load and measure dashboard and search response times.	At least 90 percent respond within 2 seconds with no corruption.	N
TC-REL-01	NF-003	Monitor the deployed test environment during planned operating time.	Measured availability meets the 99 percent target, excluding maintenance.	N
TC-SEC-01	NF-004, SR-001	Inspect protected traffic and attempt plaintext access.	TLS protects sensitive traffic and plaintext access is blocked or redirected safely.	N
TC-UX-01	NF-005	Complete common tasks on supported desktop and mobile browsers.	Pages remain understandable, responsive and keyboard usable.	N
TC-REL-02	NF-006	Inject validation and database failures during allocation or issue.	The transaction rolls back without inconsistent state.	N
TC-MAINT-01	NF-007	Review module boundaries, setup instructions and code documentation.	Core modules can be understood and changed independently.	N
TC-SEC-02	SR-002	Inspect password storage and authentication code.	Passwords use a secure one-way hash and never appear in plaintext.	N
TC-SEC-03	SR-003	Call admin, approval, allocation and issue APIs without required roles.	Every unauthorized request is rejected by the server.	N
TC-SEC-04	SR-004	Submit malformed and injection-style inputs to forms and APIs.	Input is rejected or safely handled; queries are not altered.	N
TC-SEC-05	SR-005	Review logs after approvals, allocations, issues and account changes.	Significant events include actor and timestamp.	N

15.Requirements Traceability Matrix

Requirement ID	Requirement Focus	Test Case	Status
BBMS-F-001	Register donor	TC-DON-01	N
BBMS-F-002	View and update donor	TC-DON-02	N
BBMS-F-003	Record eligibility	TC-DON-03	N
BBMS-F-004	Donation history	TC-DON-04	N
BBMS-F-010	Record donation	TC-INV-01	N
BBMS-F-011	Update inventory	TC-INV-02	N
BBMS-F-012	Display inventory	TC-INV-03	N
BBMS-F-013	Search and filter inventory	TC-INV-03	N
BBMS-F-014	Track expiry	TC-INV-04	N
BBMS-F-020	Register recipient	TC-REC-01	N
BBMS-F-021	Manage recipient records	TC-REC-01	N
BBMS-F-022	Create blood request	TC-REQ-01	N
BBMS-F-023	Validate request	TC-REQ-01	N
BBMS-F-024	Track request states	TC-REQ-02	N
BBMS-F-025	Process request by permission	TC-REQ-03	N
BBMS-F-026	Allocate compatible stock	TC-ALLOC-01	N
BBMS-F-027	Issue and fulfill blood	TC-ISSUE-01	N
BBMS-F-030	Authenticate users	TC-AUTH-01	N

15.RTM Continued

Requirement ID	Requirement Focus	Test Case	Status
BBMS-F-031	Enforce role access	TC-AUTH-02	N
BBMS-F-032	Reject invalid sessions	TC-AUTH-03	N
BBMS-F-033	Protect admin interface	TC-AUTH-02	N
BBMS-F-034	Manage users and roles	TC-ADMIN-01	N
BBMS-F-035	Provide reports	TC-ADMIN-02	N
BBMS-F-036	Record admin actions	TC-ADMIN-03	N
BBMS-NF-001	At least 90% of requests respond within 2 seconds.	TC-PERF-01	N
BBMS-NF-002	Support project workload	TC-PERF-01	N
BBMS-NF-003	99 percent planned availability	TC-REL-01	N
BBMS-NF-004	Protected communication	TC-SEC-01	N
BBMS-NF-005	Responsive understandable UI	TC-UX-01	N
BBMS-NF-006	Recoverable consistent records	TC-REL-02	N
BBMS-NF-007	Modular documented code	TC-MAINT-01	N
BBMS-SR-001	TLS for sensitive traffic	TC-SEC-01	N
BBMS-SR-002	Secure password hashing	TC-SEC-02	N
BBMS-SR-003	Authorization for protected operations	TC-SEC-03	N
BBMS-SR-004	Safe validation and database use	TC-SEC-04	N
BBMS-SR-005	Operational audit trail	TC-SEC-05	N

16.Test Metrics and Reporting

- Execution percentage and pass, fail, and blocked counts.
- Requirement coverage and high-priority requirement pass rate.
- Defects by severity, age, module, and closure status.
- 90th-percentile response time, load-test errors, and availability measurement.
- Security findings, browser or usability issues, and transaction-recovery results.

17.Approvals

Role	Name	Signature or Date
Team Representative	G S Retikka Prasad	Pending
Course Coordinator	Pradeep Kumar Kenny	Pending

18.Final Review Checklist

Checklist Item	Status
Project and team details match SRS v1.0	Yes
All 24 functional requirements are mapped	Yes
All 7 non-functional requirements are mapped	Yes
All 5 security requirements are mapped	Yes
No execution result is claimed without evidence	Yes
Actual execution and approvals	Pending